Aditya Wagh

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LinkedIn | GitHub | Portfolio



Profile

Aspiring AI researcher and final-year undergraduate in Computer Science (AIML), with hands-on experience in Generative AI, NLP, and deep learning. Passionate about contributing to cutting-edge research in LLMs and real-world intelligent systems.

Education

2022–2026
Bachelor of Technology (B.Tech) in Computer Science (Artificial Intelligence and Machine Learning)
Vishwakarma Institute of Information Technology (VIIT), Pune, India
Final CGPA: 8.32/10

2021–2022
Higher Secondary Certificate (HSC) – Science Stream
Trimurti Arts, Commerce and Science Junior College, Shrirampur, India

2019–2020
Secondary School Certificate (SSC)
St. Xavier's English Medium School, Shrirampur, India

Work Experience

Dec 2025 – Feb 2026
Research Intern
Indian Institute of Technology (IIT) Bombay – FOSSEE, Mumbai, India

Main activities and responsibilities:

- Contributed to the development and enhancement of **Osdag Bridge**, an open-source engineering software for bridge analysis and design.
- Implemented and optimized software components using **Python (PySide6)**, improving usability, performance, and maintainability.
- Collaborated with senior developers and domain experts to translate engineering requirements into scalable, production-ready solutions.

Jan 2025 – Jun 2025
Project Developer Intern
BigBuddy Foundation, India

Main activities and responsibilities:

- Designed and deployed an end-to-end **OCR-to-Excel automation pipeline** using **Tesseract OCR** and OpenAI LLMs for structured document data extraction.
- Automated extraction of **5+ structured data fields**, reducing manual data entry effort by **60–70%**.
- Developed a **MERN-stack educational web platform** supporting **100+ users**, integrating gamified modules to enhance engagement and learning outcomes.

Projects

May 2025
NeuroQuery – AI-Powered Database Query Assistant [GitHub]

- Built an NLP system that interprets voice and text queries to automatically generate SQL commands for real-time database access.
- Achieved **96% intent classification accuracy** using Scikit-learn; deployed with Flask backend and React frontend.
- Explored applications in voice-enabled BI dashboards to improve accessibility for non-technical users.

Jun 2025
NewsGenie – Personalized News Summarizer and Recommender [GitHub]

- Implemented a BART-based news summarizer that fetches live headlines and generates personalized content.
- Integrated Supabase for user authentication and preferences; deployed with Flask, React, and Hugging Face Transformers.
- Used cosine similarity and collaborative filtering to improve recommendation relevance.

Jul 2025
Blind Reader App – Assistive AI Tool [GitHub]

- Developed an accessibility-focused application that converts images into spoken descriptions for visually impaired users.
- Integrated computer vision models (BLIP/T5) with text-to-speech systems (Bark/Coqui) to generate context-aware audio outputs.
- Conducted user testing to validate inclusivity, usability, and practical effectiveness.

Digital Skills

Programming Languages: Python, Java, JavaScript, C++, SQL, HTML/CSS, Bash

AI / Machine Learning: Scikit-learn, TensorFlow, Hugging Face Transformers, OpenCV, NumPy, Pandas

Web Development Frameworks: Flask, React, Node.js, Express.js, TailwindCSS

Databases & Platforms: MongoDB, MySQL, SQLite, Firebase, Supabase

Tools & Environments: Git, Docker (basic), Jupyter Notebook, VS Code, Android Studio

Publications

- **Contextual Filler Word Detection using BERT and Prosodic Features**, 9th IEEE International Conference on Control, Communication, Computing and Automation (ICCUBEA), 2025. Published in IEEE Xplore.
- **AI Agent for Scientific Paper Analysis**, IEEE International Conference on Intelligent Signal Processing and Effective Communication Technologies (INSPECT), 2025. Published in IEEE Xplore.
- **Robust Facial Expression Recognition with VGG16: Large-Scale Training and Cross-Dataset Validation on AffectNet and FER2013**, 2nd Asia Pacific Conference on Innovation in Technology (APCIT), 2025. Accepted (to appear in IEEE Xplore).
- **Bridging the Language Gap: A Survey and Comparative Analysis of Open-Source Chatbots for Indian Languages**, International Conference on Recent Research Advances in Artificial Intelligence (ICRRAI), 2025. Accepted.

Certifications

- **Building RAG Applications Using MongoDB** – MongoDB, Aug 2025
- **IBM AI Engineering Specialization** – Coursera, Oct 2024
- **IBM Full Stack Software Developer** – Coursera, Jul 2024
- **Generative AI for Everyone** – DeepLearning.AI, Sep 2024
- **CCNA: Introduction to Networks** – Cisco, Nov 2024
- **Principles of UX/UI Design** – Meta, Oct 2024

Additional Information

- Completed the **Global Immersion Program at UCSI University, Malaysia** (Sept 2025); earned certifications in AI-Driven Cybersecurity Tools, Data Privacy in AI Applications, and AI in Ethical Decision-Making.
- Contributed to the open-source development of **Osdag Bridge**, an interdisciplinary civil engineering software for bridge analysis and design (FOSSEE Project).
- Completed the **CDAC UAV 5 day Bootcamp on AI & Drone Systems** (2025); received intensive training in AI-driven drone operations and autonomous systems.
- **National Service Scheme (NSS) Volunteer for 1 year** (2024); actively participated in community service initiatives including social awareness drives and outreach programs.

Research Interests

Natural Language Processing, Generative AI, Explainable AI, Human-AI Interaction, Responsible AI